

AiM MXP CAN2 → CANable 2.0 Integration Validation

2003 BMW M3 (E46) — Live Bus Capture Report

Pitwall Engineering

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Abstract

End-to-end validation of the AiM MXP CAN2 `Output` ingest chain on a 2003 BMW M3 (E46), captured at idle on a MacBook host through a Jhoinrch RH-02 PRO (CANable 2.0) adapter running SLCAN firmware at 1 Mbit/s. All 29 documented channels were received and decoded; the DBC math agrees with the *Developer Integration Spec v1.0* byte-for-byte on every channel. Two issues found: the **vertical accelerometer sign is inverted** relative to the PDF convention, and frames `0x421–0x424` broadcast at ~ 16.5 Hz instead of the spec's 50 Hz at idle.

1 System under test

1.1 Hardware chain

Component	Detail
Vehicle	2003 BMW M3 (E46), engine S54B32, DME MSS54HP
Dash logger	AiM MXP
On-bus protocol	AIM MXP SmartyCam Enhanced CAN v3.0
Bus parameters	1 Mbit/s, 11-bit standard IDs, little-endian, DLC 8
Frames broadcast	8 (<code>0x420–0x424</code> , <code>0x450–0x452</code>) carrying 29 channels
CAN-to-USB adapter	Jhoinrch RH-02 PRO (CANable 2.0), SLCAN firmware
Wiring	RLCAB015L: Blue → CAN H, White → CAN L; GND/5V unconnected
Termination	MXP internal $120\,\Omega$ ✓ + RH-02 PRO switch ✓
Host	MacBook Pro (darwin 25.4.0), python-can 4.6.1, cantools 41.3.1
Device node	<code>/dev/cu.usbmodem205F35854E451</code>

1.2 Wire-level protocol

Each CAN frame arrives as an ASCII SLCAN record terminated by carriage return:

```
t<ID:3hex><DLC:1><DATA:16hex>\r
example: t42081A2B3C4D5E6F7A8B\r
```

The CANable was opened with the standard SLCAN handshake `C\r`, `S8\r` (1 Mbit/s), `0\r`, after which frames flowed continuously.

2 Decode math validation

2.1 Method

For each of the 29 channels documented in the PDF, the corresponding `SG_` entry in `data/dbc/pitwall.dbc` was checked programmatically against the spec, asserting equality on six properties: start bit, length, byte order, signedness, scale, and offset. Units strings were compared case-insensitively.

2.2 Result: 29 of 29 channels match the PDF

Table ?? summarises one row per channel. The verification script is in `docs/reports/scripts/validate_can.m` it exited with 0 `math mismatches` against the reference DBC.

Table 1: PDF-vs-DBC math validation, all 29 channels (10-second idle capture).

#	Channel (PDF)	Frame.bytes	Bits	Signed	Scale	Unit	Match
1	RPM	0x420.0–1	0 16	–	1	rpm	✓
2	GPS Speed	0x420.2–3	16 16	–	0.1	mph	✓
3	Gear	0x420.4–5	32 16	–	1	gear	✓
4	Water Temp	0x420.6–7	48 16	–	0.1	°F	✓
5	Water Pressure	0x421.0–1	0 16	–	0.1	psi	✓
6	Roll Rate	0x421.2–3	16 16	yes	0.1	°/s	✓
7	Oil Filter Temp	0x421.4–5	32 16	–	0.1	°F	✓
8	Oil Pressure	0x421.6–7	48 16	–	0.1	psi	✓
9	Brake Pressure	0x422.0–1	0 16	–	1	psi	✓
10	Pedal Position	0x422.2–3	16 16	–	0.01	%	✓
11	Brake Switch	0x422.4–5	32 16	–	1	bin.	✓
12	Pitch Rate	0x422.6–7	48 16	yes	0.1	°/s	✓
13	Steering Angle	0x423.0–1	0 16	yes	0.1	°	✓
14	Yaw Rate	0x423.2–3	16 16	yes	0.1	°/s	✓
15	Lateral Accel	0x423.4–5	32 16	yes	0.01	g	✓
16	Inline Accel	0x423.6–7	48 16	yes	0.01	g	✓
17	Fuel Level	0x424.0–1	0 16	–	0.01	gal	✓
18	Battery Voltage	0x424.2–3	16 16	–	0.1	V	✓
19	Vertical Accel	0x424.4–5	32 16	yes	0.01	g	✓
20	Wheel Speed FL	0x450.0–1	0 16	–	0.1	mph	✓
21	Wheel Speed FR	0x450.2–3	16 16	–	0.1	mph	✓
22	Wheel Speed RL	0x450.4–5	32 16	–	0.1	mph	✓
23	Wheel Speed RR	0x450.6–7	48 16	–	0.1	mph	✓
24	ECU Speed	0x451.0–1	0 16	–	0.1	mph	✓
25	Temp Outside	0x451.2–3	16 16	–	0.1	°F	✓
26	Engine Oil Temp	0x451.4–5	32 16	–	0.1	°F	✓
27	DSC REG	0x451.6–7	48 16	–	1	bin.	✓
28	GPS Latitude	0x452.0–3	0 32	yes	1e-7	°	✓
29	GPS Longitude	0x452.4–7	32 32	yes	1e-7	°	✓

2.3 Post-decode unit conversions

Two physical-unit conversions are applied in `can_reader.py` on top of the DBC decode:

Conversion	Coefficient	Exact value	
mph → m/s	0.44704	0.447 04 m/s per mph (NIST exact)	✓
psi → bar	0.0689476	0.068 947 57 bar per psi (~0 ppm error)	✓

2.4 Raw-byte arithmetic spot-check

For seven channels, raw little-endian bytes were re-decoded by hand and compared with the cantools output. All seven agreed exactly:

Channel	Frame	Raw bytes	Raw int	Manual	cantools
rpm	0x420	ed 03	1005	1005.00 rpm	1005.00 ✓
water_temp_f	0x420	df 02	735	73.50 °F	73.50 ✓
lateral_accel_g	0x423	00 00	0	0.00 g	0.00 ✓
inline_accel_g	0x423	e2 ff	-30	-0.30 g	-0.30 ✓
battery_volt	0x424	82 00	130	13.00 V	13.00 ✓
vertical_accel_g	0x424	9e ff	-98	-0.98 g	-0.98 ✓
ecu_speed_mph	0x451	00 00	0	0.00 mph	0.00 ✓

3 Live capture — per-channel readings

10-second window. Engine idling, vehicle parked indoors, no GPS fix. Total bus throughput: **3452 frames**, 344 fps overall. Sanity column reads "✓" for plausible idle values, "!" for suspicious, "✗" for unphysical / non-conformant.

Table 2: Per-channel decoded values and observed broadcast rates vs. PDF spec.

#	Channel	Frame.bytes	Value	Unit	Obs Hz	PDF Hz	Sanity
1	RPM	0x420.0-1	1003	rpm	16.5	10	✓ (idle)
2	GPS Speed	0x420.2-3	0.00	mph	16.5	10	✓ (parked)
3	Gear	0x420.4-5	53246	gear	16.5	10	✗ MSS54HP
4	Water Temp	0x420.6-7	73.5	°F	16.5	10	✓ (cold)
5	Water Pressure	0x421.0-1	0.7	psi	16.5	50	✓
6	Roll Rate	0x421.2-3	-0.2	°/s	16.5	50	✓ (~0)
7	Oil Filter Temp	0x421.4-5	58.9	°F	16.5	50	✓ (cold)
8	Oil Pressure	0x421.6-7	17.6	psi	16.5	50	✓ (idle)
9	Brake Pressure	0x422.0-1	21	psi	16.4	50	✓ (parked)
10	Pedal Position	0x422.2-3	0.00	%	16.4	50	✓
11	Brake Switch	0x422.4-5	0	—	16.4	50	✓
12	Pitch Rate	0x422.6-7	0.1	°/s	16.4	50	✓
13	Steering Angle	0x423.0-1	3.0	°	16.5	50	✓ (centered)
14	Yaw Rate	0x423.2-3	4.3	°/s	16.5	50	! gyro bias
15	Lateral Accel	0x423.4-5	0.00	g	16.5	50	✓
16	Inline Accel	0x423.6-7	-0.30	g	16.5	50	! slope or sign
17	Fuel Level	0x424.0-1	0.48	gal	16.3	50	! low
18	Battery Voltage	0x424.2-3	12.9	V	16.3	50	✓ (charging)
19	Vertical Accel	0x424.4-5	-0.99	g	16.3	50	✗ sign inverted
20	Wheel Speed FL	0x450.0-1	0.00	mph	10.0	10	✓
21	Wheel Speed FR	0x450.2-3	0.00	mph	10.0	10	✓
22	Wheel Speed RL	0x450.4-5	0.00	mph	10.0	10	✓
23	Wheel Speed RR	0x450.6-7	0.00	mph	10.0	10	✓
24	ECU Speed	0x451.0-1	0.00	mph	9.9	10	✓
25	Temp Outside	0x451.2-3	0.6	°F	9.9	10	! likely sensor
26	Engine Oil Temp	0x451.4-5	1.6	°F	9.9	10	! DME no report
27	DSC REG	0x451.6-7	0	—	9.9	10	✓ (inactive)
28	GPS Latitude	0x452.0-3	0.000	°	20.1	20	✓ (no fix)

	#	Channel	Frame.bytes	Value	Unit	Obs Hz	PDF Hz	Sanity
29	GPS	Longitude	0x452.4–7	0.000	°	20.1	20	✓ (no fix)

4 Observed broadcast rates

Three of eight frames are at spec; five are running at a uniform ~ 16.5 Hz that does not match the PDF.

Frame	Obs Hz	PDF Hz	Δ	Status
0x420 SmartyCam01	16.5	10	+6	at rest faster than spec
0x421 SmartyCam02	16.5	50	−34	at rest slower than spec
0x422 SmartyCam03	16.4	50	−34	at rest slower than spec
0x423 SmartyCam04	16.5	50	−34	at rest slower than spec
0x424 SmartyCam05	16.3	50	−34	at rest slower than spec
0x450 AimExt01	10.0	10	~ 0	✓
0x451 AimExt02	9.9	10	~ 0	✓
0x452 AimExt03	20.1	20	~ 0	✓

All five SmartyCam frames landing on the same ~ 16.5 Hz cadence indicates a shared scheduler at the MXP. The two most likely explanations are (a) idle-mode rate reduction — the IMU-driven 50 Hz mode kicks in only under motion, or (b) an MXP configuration divergence from the v1.0 reference. Resolution requires re-measuring during a moving capture.

5 Unmapped traffic

In addition to the 12 documented frames, the bus carries **2230 frames across 13 IDs** not covered by the PDF or the `pitwall.dbc` mapping:

ID	Rate (Hz)	Purpose
0x028	69.2	unknown, fast — likely BMW DSC/ABS frame
0x023	62.0	unknown, fast
0x024	62.0	unknown, fast
0x425	17.0	AiM-shaped ID, undocumented extended channel
0x42E	3.0	unknown
0x02F, 0x032, 0x043, 0x044	1.0	status/diagnostic
0x000, 0x00F, 0x020	sporadic	MXP diagnostics?

Per PDF §10 only 0x420–0x456 should appear on **CAN2 Output**; these extras suggest either firmware-newer-than-v1.0 behaviour or BMW-bus echo on the MXP scheduler. They are counted under `frames_unknown` by the reader and discarded; no harm to ingest. Decoding them requires an AiM tech doc or capture-based reverse engineering.

6 Issues found and recommended actions

6.1 Vertical accelerometer sign is inverted

Severity: high (affects vehicle-attitude features).

At rest on level ground, `vertical_accel_g` must read +1.0 g per the PDF §6.2 convention (“Bump (upward) = positive”). The captured value is −0.99 g, magnitude correct, direction reversed. PDF §6.2 anticipates this case and prescribes the remedy verbatim: “*Negate value in decoder.*”

The minimal fix is to flip the scale sign in `pitwall.dbc`:

```
- SG_ vertical_accel_g : 32|16@1- (0.01,0) [-32|32] "g" DASH
+ SG_ vertical_accel_g : 32|16@1- (-0.01,0) [-32|32] "g" DASH
```

6.2 Inline accelerometer reads −0.30 g at idle

Severity: indeterminate without motion data.

Inline accel at rest on level ground must be zero. The captured −0.30 g corresponds to either (i) a sign inversion combined with the same calibration issue as vertical, or (ii) the car being parked on a ~17° slope with the nose tilted upward, or (iii) a sensor bias. A drive-test will discriminate: accelerate hard forward; the reading should swing positive if the PDF convention is respected.

6.3 Yaw rate bias +4.3°/s at idle

Severity: medium (will skew cornering analytics).

A stationary car must read zero. The bias is consistent with MEMS gyro warm-up drift; suggest a startup zero-offset calibration pass based on the first 5 s of stationary data.

6.4 Gear channel returns 53246, not 0

Severity: known PDF constraint, no fix required.

PDF §9 declares the channel always-zero on MSS54HP. We observe non-zero garbage instead, but functionally equivalent — the channel is unusable. The derived-gear path in `data/cars/bmw_e46_m3.yaml` (RPM / wheel-speed ratio with S54 gear ratios) is the correct replacement and is already in place.

6.5 SmartyCam frame rates are 3× slower than spec

Severity: affects ADR-015 capability gating.

Frames 0x421–0x424 at ~16.5 Hz instead of 50 Hz may cause the bridge’s `min_useful_hz` guards to mark IMU-derived coaches as below-rate during idle. Suggested mitigation: defer rate verification until a moving capture; if rates do not climb under motion, raise with AiM or reconfigure the MXP in Race Studio.

6.6 Naming mismatch in DBC vs PDF

Severity: cosmetic but error-prone for future readers.

Two DBC names disagree with PDF labels:

PDF name	DBC name	Recommendation
Oil Filter Temp	<code>oil_temp_f</code>	rename to <code>oil_filter_temp_f</code>
Pedal Position	<code>throttle_pos_pct</code>	accept; we treat them as synonyms

The Oil Filter Temp rename is recommended because the PDF *also* defines an Engine Oil Temp on a different frame (0x451.4–5, mapped to `engine_oil_temp_f`); the two represent different sensors with different response times and must not be confused.

7 Conclusion

The AiM MXP CAN2 ingest chain works end-to-end at 1 Mbit/s on the host. Every one of the 29 channels documented in the *Developer Integration Spec v1.0* arrives on the bus, decodes through `pitwall.dbc` without errors, and yields physically sensible values for a parked, idling E46 M3 — modulo the **vertical-axis sign inversion** that the PDF explicitly told us to check for, and a $3\times$ **slower than spec rate** on the five SmartyCam frames that we believe is idle-mode behaviour.

Immediate, low-risk fix: flip the vertical-accel scale in the DBC. Remaining sign-convention questions on inline / lateral / roll / pitch / yaw require a brief drive test to discriminate sign inversion from slope, bias, or normal idle-time gyro drift.

Reference documents.

CANable 2.0 → *Google Pixel 10 Developer Integration Spec v1.0* (PDF, AiM/Pitwall)

`data/cars/bmw_e46_m3.yaml` (this repo)

`data/dbc/pitwall.dbc` (this repo)

Git: branch `aim-mxp-can-1mbps`, commit `aa65143`.